

NETWORKS AND COMPLEXITY

Solution 22-5

This is an example solution from the forthcoming book Networks and Complexity.

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 22.5: Product Formula [Lvl. 3]

Consider the Kronecker product

$$\mathbf{J} = \mathbf{A} \otimes \mathbf{B}. \quad (1)$$

Find a formula that relates the eigenvalues of $\mathbf{J}$ to the eigenvalues of $\mathbf{A}$ and $\mathbf{B}$.

Solution

We use the Ansatz

$$\mathbf{v} = \mathbf{a} \otimes \mathbf{b}. \quad (2)$$

We now show that (under suitable assumptions) this is an eigenvector of $\mathbf{J}$. We write

$$\mathbf{J}\mathbf{v} = (\mathbf{A} \otimes \mathbf{B})(\mathbf{a} \otimes \mathbf{b}) \quad (3)$$

$$= \mathbf{A}\mathbf{a} \otimes \mathbf{B}\mathbf{b} \quad (4)$$

$$= \alpha\mathbf{a} \otimes \beta\mathbf{b} \quad (5)$$

$$= \alpha\beta(\mathbf{a} \otimes \mathbf{b}) \quad (6)$$

$$= \alpha\beta\mathbf{v} \quad (7)$$

Where we had to assume that $\mathbf{a}$ is an eigenvector of $\mathbf{A}$ with eigenvalue α (i.e. $\mathbf{A}\mathbf{a} = \alpha\mathbf{a}$) and similarly $\mathbf{B}\mathbf{b} = \beta\mathbf{b}$.

The calculation above shows that if $\mathbf{J}$ is a Kronecker product of 2 matrices and we know the eigenvectors of these matrices, then the Kronecker product of a pair of eigenvectors is an eigenvector of $\mathbf{J}$ and the corresponding eigenvalue is the product of the two eigenvalues.

We can see that the number of eigenvalues we can construct in this way is

$$\dim(\mathbf{A}) \cdot \dim(\mathbf{B}) = \dim(\mathbf{J}) \quad (8)$$

. Hence, we can find all eigenvalues and eigenvectors of $\mathbf{J}$ in this way.